

Dynamic Wrinkling on Coarsely Meshed Cloth

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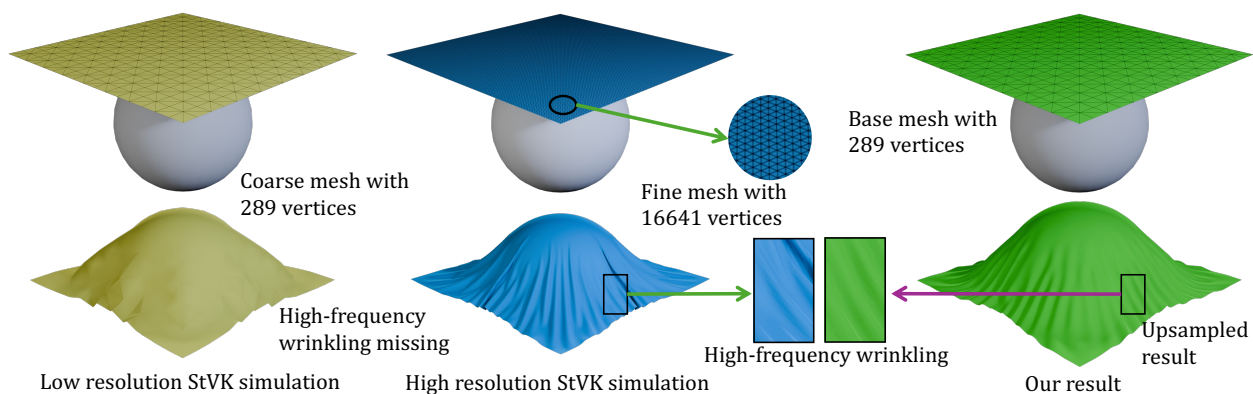


Figure 1: We present a simulation pipeline that extends tension-field wrinkles (TFW) [CCK*21] for dynamic wrinkling on coarse meshes. This figure compares dynamic simulations of a cloth piece falling onto a sphere under gravity. Top: initial state and mesh discretization. Bottom: an intermediate simulation frame taken as the cloth falls onto the sphere.

Abstract

We present a technique for simulating detailed cloth dynamics on coarse meshes at interactive rates. Building on the wrinkle parameterization introduced by [CCK*21], we introduce a dynamics model for a cloth sheet represented as a superposition of a base surface and a wrinkle distribution parameterized by spatially varying amplitude and frequency. Our model incorporates two-way coupling of the base surface and the wrinkle parameters, allowing the base deformation to drive emergence of wrinkles and permitting compression of the base surface in turn. To deform the mesh using the computed wrinkle parameters, we also introduce a simple phase reconstruction strategy that produces dynamically evolving, temporally coherent wrinkles on the simulated mesh.

CCS Concepts

• **Computing methodologies** → **Physical simulation**;

1. Introduction

Cloth simulation techniques in computer graphics have seen dramatic improvements over the last several decades. Offline simulations of garments with many millions of vertices can model fabric at sub-millimeter scales [Wan21] and on the level of individual yarns [KJM08; CLMO14]. However, for interactive applications, simulation quality is still limited by mesh resolution, and modeling clothing with realistic details such as wrinkles remains a challenge.

The fundamental challenge in the interactive setting is that the cost of simulation rises rapidly with mesh resolution. Discretizing

the garment at a fine enough resolution to capture all visually important details such as emergent wrinkles is infeasible to simulate at interactive rates. On the other hand, simulations on coarser meshes suffer not just from a lack of fine details, but also visually objectionable artifacts arising from the inability to resolve sub-element deformations. In essence, coarse discretizations smooth out the deformation resulting in loss of high frequency details, whereas fine meshes increase the problem stiffness requiring more expensive solvers.

Wrinkles play an important role in providing visual plausibility for cloth simulations. Recently, Chen et al. [CCK*21] have

introduced a physics-based technique for modeling fine wrinkles on coarsely meshed sheets, by first simulating the deformation of the coarse mesh and then computing the spatially varying wrinkle parameters (amplitude and phase). Their technique yields impressive agreement with reference high-resolution simulations in many cases, but only works for static drapes and does not support animation. Their follow-up work [CKSV23] allows animating wrinkles evolving over time through a spacetime optimization, which is not suitable for interactive applications.

In this work, we build on the wrinkle representation of Chen et al. [CCK*21] to provide a physically consistent method for dynamic cloth simulation with fine wrinkles on coarse cloth meshes. Our approach supports two-way coupling of the wrinkle parameters with the coarse base mesh, allowing the emerging wrinkles to affect the mechanical behavior of the base mesh and reducing locking artifacts. We also provide a fast method to synthesize animated wrinkles in a temporally coherent manner.

Our main contributions are:

1. We present the first physics-based model for the dynamics of an elastic sheet represented as a mid-surface coupled with wrinkle amplitudes and frequencies.
2. We design a simple spatial discretization and time stepping scheme for simulation of such wrinkle-augmented sheets.
3. We present a fast method for temporally coherent reconstruction of phases from the frequency wrinkle field.

2. Related work

Many cloth simulation techniques have appeared in graphics literature, from mass-spring models [Bre93; Pro95; CK02] to finite elements [TWS06; NSO12] and yarn-level simulation [KJM08; CLMO14]. Notable recent advances enable differentiable simulation [LLK19; LDW*22] and improve collision handling performance [LLL*24]. Neural networks have also been applied for cloth simulation [SOC22; BME22; GBH23; KHTG24; LWK*24], although most are restricted to garments on human characters and typically require garment- and body-specific training.

In this work, we focus on mesh-based cloth simulation techniques suitable for general-purpose interactive simulation. The buckling instability in cloth naturally produces wrinkles and folds at small scales, which are essential for visual realism. We distinguish these from the persistent wrinkles that remain after the cloth is strongly deformed, which arise due to internal friction [MTB*13; GYSW25] and are outside the scope of our work. At lower resolutions, the simulation mesh is unable to resolve emergent buckling details, causing the dynamics to deviate significantly from the desired behavior [ZJK25], and sometimes resulting in severe simulation artifacts [KNO15]. These problems can be mitigated by refining the mesh to provide enough degrees of freedom for wrinkles to appear [NSO12; ZWL*25], but this increases the cost of the simulation and is not feasible in interactive applications. Alternatively, artifacts such as locking can also be mitigated by lowering the stiffness of the cloth, but as pointed out by [GHF*07] this leads to an objectionably stretchy appearance.

A promising approach to overcoming these challenges is to conceptually separate the fine-scale details such as wrinkles and folds

from the overall low-frequency “base” deformation of the sheet, allowing a coarse simulation mesh to be responsible only for the latter, and then augment the simulation with wrinkle details as a postprocess. The seminal work of Rohmer et al. [RPC*10] traced wrinkle curves along the simulated sheet using the stretch tensor and used them to deform the surface. Building on this approach, Gillette et al. [GPV*15] described a careful stretch tensor estimation process to add wrinkles to extremely low-poly cloth animation with no reference shape. Other works designing procedural models for wrinkles include [ZH20; MSYA21]. Alternatively, the wrinkle patterns arising from base mesh deformation can be learned from precomputed data. For a specific garment/character pair, Wang et al. [WHRO10] precomputed a database containing the wrinkle patterns for different character poses, and applied them to a coarse simulation mesh at runtime. Neural techniques for this task have also been proposed recently [ZWCM21; ZL24], showing ability to generalize to novel garments beyond the ones in the training set. However, existing learned models do not account for the temporal dynamics of wrinkles, and as a result the synthesized results tend to exhibit fairly static wrinkles that fade in and out over time rather than sliding, merging, and splitting realistically. In addition, all previous techniques discussed in this paragraph treat only the wrinkle behavior as driven by the base deformation, but do not consider the effect of wrinkles on the mechanical response of the base deformation itself.

Along these lines, we also note several techniques for augmenting coarse models with wrinkles by performing additional simulation at high resolution, coupled with the coarse deformation [BMWG07; MC10; RK13; CXY*23]. However, these incur an additional computational expense for the high-resolution simulation. The recently introduced framework of progressive dynamics [ZJK24; ZJK25] allows the spatial resolution of the simulation to be gradually increased while preserving the overall motion of the original coarse simulation. However, in this approach, the elastic behaviour of the coarse mesh is computed through prolongation, so it does not aim to model wrinkles finer than the mesh resolution which is the goal of our work.

Wrinkle mechanics has been extensively studied in physics. For instance, Cerda and Mahadevan [CM03] show the relationship between wrinkle wavelengths and amplitudes for a stretched thin sheet, assuming that wrinkles form only in the direction perpendicular to the base shell. Healey et al. [HLC13] give physical bounds for the shape of a thin sheet to form wrinkle patterns. They show that the Foppl-von Karman theory fails to provide stable bifurcation solution for shells with large mid-surface strain, and use a pure Green strain model to model larger non-linear strain response in the wrinkled regions. Aharoni et al. [ATA*17] parameterize wrinkling of skin-like thin shells using sinusoids and derive the relationship between compressive strain, wrinkle wavelength and amplitudes. While these methods describe the overall relationship between material properties, sheet geometry, and wrinkling behavior, they cannot directly be applied to dynamic simulations where both the base mesh deformation and the wrinkle directions are varying over time.

Recently Chen et al. [CCK*21] proposed tension-field wrinkles (TFW), a physics-based method for modeling wrinkle behavior without requiring an additional fine-resolution mesh. Their model

is based on the wrinkle mesh [ATA*17], in which the wrinkled sheet is modeled as a coarse base mesh augmented with wrinkle-correction terms parameterized by amplitude and spatial frequency. They use a two-stage quasi-static simulation pipeline where, first, the base mesh is simulated to obtain the correct static shape. Then, wrinkle parameters are obtained by optimizing the elastic energy of the wrinkle mesh. A major limitation of Chen et al.'s [CCK*21] model is the missing coupling between wrinkling and coarse mesh deformation. For instance, a highly wrinkled region might cause the base mesh to expand due to bending and collision forces between adjacent wrinkles, but their model lacks this coupling. Another limitation of their work is that it only computes static drapes and does not support animation.

The temporal evolution of the wrinkle mesh has been explored in Chen et al. [CKSV23], which solves an intermediate-value problem given a fixed base surface and the wrinkle parameters at the temporal endpoints. To this end, they formulate a Dirichlet energy that penalizes the bending strain of the wrinkle mesh between the endpoints. Their wrinkle mesh model comprises amplitudes, wrinkle frequencies, and wrinkle phase, respectively. This formulation deviates from the wrinkle mesh used previously [CCK*21], where only wrinkle frequencies are considered as physical parameters in defining the elastic energy, which are then used to compute a compatible phase for rendering. Thus, a soft-compatibility term is introduced as a constraint to the Dirichlet energy for obtaining temporally consistent wrinkle parameters. Their formulation provides smooth interpolation steps that look visually plausible; however, it does not allow forward simulation. We can use the quasi-static solution of Chen et al. [CCK*21] to obtain the interpolation endpoints between the starting and ending time frames, and then use the interpolation to obtain the dynamics of the wrinkle mesh. Unfortunately, the method would fail to reproduce physically valid dynamics because the inertia of the wrinkle parameters is ignored. Moreover, the simulations would look highly damped. Furthermore, optimizing the Dirichlet energy at each simulation step is computationally expensive and cannot be used for real-time simulations.

3. Background

Our work builds on the approach of Chen et al. [CCK*21] for quasi-static modeling of fine wrinkles on coarsely meshed thin shells. For completeness, we summarize the salient aspects of their method here.

The cloth is treated as an elastic thin shell of thickness h . Using the Kirchhoff-Love assumption, the deformed state of the shell can be represented in terms of its mid-surface $\mathbf{r} : \Omega \rightarrow \mathbb{R}^3$, where $\Omega \subset \mathbb{R}^2$ is the reference domain. To efficiently represent the wrinkled shape of the shell, they decompose the deformation into a smooth *base surface* $\mathbf{r}_b$, which captures large-scale cloth deformation, and a sinusoidal displacement field with spatially varying amplitude a and phase ϕ which adds fine-scale wrinkle details:

$$\mathbf{r}(u, v) = \mathbf{r}_b(u, v) + d\mathbf{r}_b(u, v)\mathbf{v}_t(u, v) + a\cos(\phi)\mathbf{n}(u, v). \quad (1)$$

Here, $\mathbf{n}$ is the normal vector to the base surface $\mathbf{r}_b$, and the operator d denotes the Jacobian with respect to the reference coordinates (u, v) . We refer to the amplitude a and phase ϕ collectively as the *wrinkle parameters*. The term $d\mathbf{r}_b(u, v)\mathbf{v}_t(u, v) = d\mathbf{r}_b(\mathbf{v}_1 \sin \phi +$

$\mathbf{v}_2 \sin 2\phi)$ adds in-plane tangential displacements which improve the appearance of large-amplitude wrinkles. The formulation in (1) is effective at representing wrinkles that are spatially coherent and have slowly varying size and orientation, which typically arise from draping of loose garments. It does not model more complex wrinkle patterns that cannot be approximated by a single sinusoid, such as the diamond buckling pattern seen in the presence of both compression and bending (Fig. 5 top).

Since all the variables in the above model are assumed to be slowly varying, they can be discretized on a coarsely triangulated mesh. The mesh geometry represents only the base surface, while the wrinkle parameters a, ϕ are stored at the mesh vertices. To perform quasistatic simulation using this model, Chen et al. begin with the elastic energy of the wrinkled shell using the St. Venant-Kirchhoff elasticity model with Lamé parameters α and β ,

$$E = \frac{1}{2} \int_{\Omega} \left(\frac{h}{4} W_s + \frac{h^3}{12} W_b \right) \sqrt{\det(\mathbf{I}_u)} du dv, \quad (2)$$

$$W_s = \|\mathbf{I}_u^{-1}(\mathbf{I} - \mathbf{I}_u)\|_{SV}^2, \quad (3)$$

$$W_b = \|\mathbf{I}_u^{-1}(\mathbf{\Pi} - \mathbf{\Pi}_u)\|_{SV}^2 \quad (4)$$

where $\mathbf{I} = d\mathbf{r}^T d\mathbf{r}$ and $\mathbf{\Pi} = -d\mathbf{r}^T d\mathbf{n}$ are the first and second fundamental forms, and $\|\mathbf{M}\|_{SV}^2 = \frac{\alpha}{2} \text{tr}^2(\mathbf{M}) + \beta \text{tr}(\mathbf{M}^2)$. Finally, $\mathbf{I}_u$ and $\mathbf{\Pi}_u$ denote the fundamental forms of the rest geometry.

Substituting the fundamental forms (1) into the elastic potential energy (2)-(4) leads to an expression with many terms involving trigonometric functions in ϕ . The energy can then be simplified by *coarse-graining* [ATA*17], that is, averaging the energy over a neighbourhood of size comparable to one wrinkle wavelength. This eliminates oscillatory terms like $\sin \phi$, and simplifies other terms like $\sin^2 \phi$ to constants. The resulting energy is of the form

$$E = \int_{\Omega} \left(\frac{h}{4} (W_s^1 + W_s^2 + W_s^3 + W_s^4) + \frac{h^3}{12} (W_b^1 + W_b^2) \right) \sqrt{\det \mathbf{I}_u} du dv. \quad (5)$$

The most important term here is

$$W_s^1 = \left\| \mathbf{I}_u^{-1} \left(\mathbf{I}_b - \mathbf{I}_u + \frac{1}{2} d a d a^T + \frac{1}{2} a^2 d \phi d \phi^T \right) \right\|_{SV}^2, \quad (6)$$

which couples the base deformation to the wrinkle parameters. In particular, when the base surface is compressed (resulting in negative eigenvalues of $\mathbf{I}_b - \mathbf{I}_u$), the energy can be decreased by increasing the wrinkle amplitude a and spatial frequency $d\phi$. It is this effect that drives the emergence of wrinkles. As for the remaining terms, W_b^1 is identical to W_b in (4), while $W_s^2, W_s^3, W_s^4, W_b^2$ penalize the additional stretching and bending caused by wrinkles; we refer the reader to [CCK*21] for their exact expressions.

The static simulation pipeline proposed by Chen et al. [CCK*21] proceeds as follows. First, the deformation of the base surface is estimated using *tension field theory* (TFT), using a relaxed StVK model in which compression resistance is eliminated [MTMP20], along with a quadratic bending model [BWH*06]. Next, the wrinkle parameters are computed by minimizing the wrinkle mesh energy E (5). However, since the elastic energy depends on wrin-

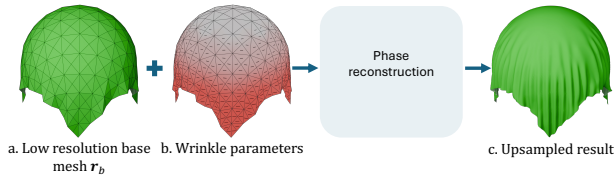


Figure 2: An overview of our method. First, we simulate the base mesh and wrinkle parameters, which are then used to generate the wrinkled mesh. Wrinkle amplitudes are visualized as colors and frequencies as lines.

kle frequencies $d\phi$ rather than on phase values ϕ directly, the optimization is performed in two stages: first the amplitudes a and frequencies $d\phi$ are obtained, then an expensive mixed-integer optimization is performed to obtain a phase field ϕ compatible with $d\phi$. This technique yields impressive agreement with reference high-resolution simulations in many cases, but the method is limited to static scenes. One could in principle use the wrinkle interpolation scheme proposed by [CKSV23] to imitate dynamics, however that does not correspond to true physical simulation.

Thus, our motivation in this work is to extend the statics formulation of wrinkle shell simulation to full dynamics, with time-varying amplitudes and phases on a moving, deforming base mesh. This requires characterizing not just the elastic potential but also the kinetic energy associated with the wrinkle parameters. In addition, we require a method to compute the phase distribution on the mesh that is not just consistent with the desired frequencies, but also exhibits temporally coherent dynamics.

4. Dynamic wrinkling

We address the issues discussed above by deriving the Lagrangian mechanics equations for the wrinkle mesh and propose a framework that simultaneously updates the base mesh and wrinkle parameters. To this end, a two-step simulation framework is devised: we simulate dynamics via Lagrangian mechanics for the wrinkle mesh to compute the base surface and the wrinkle amplitudes and frequencies, then use a variational formulation to perform temporally coherent reconstruction of the phase field from the spatial frequency field. In particular, we derive the dynamics equations for the base mesh deformation $\mathbf{r}_b$ and the wrinkle amplitude a , discretized on mesh vertices, while we treat the frequency $d\phi$ as a quasi-static variable. We also introduce a heuristic approach to associate some dynamical inertia to wrinkle phases; this allows wrinkles to slide across the mesh over time, which was not possible in earlier approaches. In summary, our method uses the true dynamics of amplitudes and base mesh deformations, augmented with a heuristic dynamic model for wrinkle phases.

4.1. Dynamic simulation

We obtain the equations of motion for the wrinkle mesh using a Lagrangian mechanics formulation, which requires defining the kinetic and potential energies associated with the degrees of freedom. We use the elastic energy formulation in (5) to model the potential

energy of the deformed wrinkle mesh. The kinetic energy is obtained by computing $T = \frac{1}{2} \dot{\mathbf{r}}^T \mathbf{M} \dot{\mathbf{r}}$, where $\mathbf{M}$ is the mass matrix. For simplicity, we drop the tangential correction term from (1), instead using only $\mathbf{r} \approx \mathbf{r}_b + a \cos(\phi) \mathbf{n}_b$. Then, T is given by

$$\begin{aligned} \frac{1}{2} \dot{\mathbf{r}}^T \mathbf{M} \dot{\mathbf{r}} = & \frac{1}{2} (\dot{\mathbf{r}}_b^T \mathbf{M} \dot{\mathbf{r}}_b) + \frac{1}{2} (\dot{\mathbf{r}}_b^T \mathbf{M} \dot{\mathbf{n}}_b) [2a \cos \phi - 2a \dot{\phi} \sin \phi] \\ & + \frac{1}{2} (\dot{\mathbf{r}}_b^T \mathbf{M} \dot{\mathbf{n}}_b) [2a \cos \phi] + \frac{1}{2} (\dot{\mathbf{n}}_b^T \mathbf{M} \dot{\mathbf{n}}_b) [a^2 \cos^2 \phi] \\ & + \frac{1}{2} (\dot{\mathbf{n}}_b^T \mathbf{M} \dot{\mathbf{n}}_b) [2a \dot{a} \cos^2 \phi - a^2 \dot{\phi} \sin 2\phi] \\ & + \frac{1}{2} (\dot{\mathbf{n}}_b^T \mathbf{M} \dot{\mathbf{n}}_b) [a^2 \cos^2 \phi + a^2 \dot{\phi}^2 \sin^2 \phi - a \dot{a} \dot{\phi} \sin 2\phi]. \end{aligned} \quad (7)$$

We treat ϕ as a quasi-static variable, and neglect its time derivatives in the above equation. Coarse-graining then gives

$$T = \frac{1}{2} \left(\dot{\mathbf{r}}_b^T \mathbf{M} \dot{\mathbf{r}}_b + \frac{a^2}{2} \dot{\mathbf{n}}_b^T \mathbf{M} \dot{\mathbf{n}}_b + \frac{\dot{a}^2}{2} \mathbf{n}_b^T \mathbf{M} \mathbf{n}_b + \frac{a^2}{2} \dot{\phi}^2 \mathbf{n}_b^T \mathbf{M} \mathbf{n}_b \right). \quad (8)$$

Finally, the Lagrangian mechanics equations of motion $\frac{d}{dt} (\partial T / \partial \dot{\mathbf{q}}) + \partial E / \partial \mathbf{q} = 0$ lead to the following governing equations for our framework:

$$\mathbf{M} \begin{pmatrix} \ddot{\mathbf{r}}_b \\ 0.5 \ddot{a} \end{pmatrix} = - \begin{pmatrix} \partial E / \partial \mathbf{r}_b \\ \partial E / \partial a \end{pmatrix} \quad (9)$$

$$d\phi = \arg \min_{d\phi} E(a, d\phi, \mathbf{r}_b) \quad (10)$$

4.2. Constraints on wrinkle parameters

Simply applying the above equations of motion leads to some visually objectionable artifacts, as we discuss below. To address these, we impose some additional constraints on the wrinkle parameters. First, we constrain the amplitudes to be nonnegative, since E is an even function of a and negative amplitudes are not physically meaningful.

Second, deformable bodies with the StVK model are prone to element collapse and inversion due to the non-convexity of the strain energy function in compressive regimes [SB12; KE20]. This issue is aggravated by the wrinkle model (6), which further reduces resistance to compression by increasing the wrinkle parameters to compensate. We propose a physical limit on compression of thin shells to restrict wrinkling. Consider two nearby points on a highly wrinkled sheet, separated by a reference-space vector $\mathbf{v}$. There are $n = |d\phi \cdot \mathbf{v}| / 2\pi$ wrinkles between the points, which have a total thickness of $2nh$. To avoid self-collisions between the wrinkles, this quantity should be less than the deformed distance between the points, $\|\mathbf{F}_b \mathbf{v}\|$ where $\mathbf{F}_b$ is the deformation gradient of the base surface. This constraint is enforced by requiring that the matrix $\mathbf{I}_b - (h/\pi)^2 d\phi d\phi^T$ is positive semidefinite.

Finally, we want the wrinkle parameters to capture only fine-scale buckling patterns, while leaving low-frequency deformations to the base mesh. In particular, we require that the wavelength of wrinkles is at most 4 times the length of a typical mesh edge, since wrinkles with longer wavelength can be easily captured by the base mesh. This is encoded in the constraint $\|d\phi\| \geq \delta\pi/2$, where δ is the average edge length of the mesh.

4.3. Discretization and Time Integration

We represent the base surface using a triangular finite element mesh, whose nodal positions are stored as a vector $\mathbf{x} \in \mathbb{R}^{3n}$. The wrinkle amplitudes and phases are also discretized on the mesh vertices, denoted by the vectors $\mathbf{a}, \phi \in \mathbb{R}^n$. The equations of motion do not involve the actual phase values but only their gradients, i.e. the wrinkle frequencies $d\phi$, which we discretize on faces. This allows us to accelerate the optimization problem in (10), since the frequency $d\phi$ on each face can be optimized independently. This is in contrast to the approach of Chen et al. [CCK*21], where phase gradients are stored as edge one-forms, and an integrability constraint is imposed that requires a global optimization. In our method, we drop the integrability constraint, since buckling patterns in cloth may have singularities where wrinkles split and merge. This provides a significant speedup for dynamic simulations.

The configuration of the system is described by the generalized coordinates $\mathbf{q} \in \mathbb{R}^{4n+2f}$. This state vector comprises the discretized vertex positions of the base mesh, the nodal wrinkle amplitudes, and the per-face wrinkle frequencies. Combining these contributions, we obtain the total system dimension:

$$\mathbf{q} = [\mathbf{a} \quad d\phi \quad \mathbf{x}]^T \in \mathbb{R}^{4n+2f} \quad (11)$$

where n and f denote the number of vertices and faces in the mesh, respectively. At each time step, we update the vertex positions $\mathbf{r}$ and amplitudes a by integrating (9) using symplectic Euler. Then, wrinkle frequencies are obtained by optimizing equation (10) independently on each face using L-BFGS with a tolerance of 10^{-6} and a maximum of 100 iterations per time step. Finally, all the state variables $\mathbf{q}$ are projected to satisfy the constraints from Section 4.2, in the style of position-based dynamics.

4.4. Wrinkle phase reconstruction

The updated state vector from the previous section provides updated positions, amplitudes, and frequencies for the end of the time step. Then, we must recover the phases ϕ per vertex to obtain the final deformed state (1). Obtaining a valid set of compatible phase values at each frame and ensuring that the deformations are temporally coherent over time is a challenging problem. In this section, we describe our method for reconstructing a compatible and time-coherent phase field from the frequency field $d\phi$.

As we do not enforce the integrability constraint while solving for $d\phi$ in the static update step, the phase gradients obtained in the simulation update are almost never integrable. Thus, we follow [CKSV23]'s idea of finding an approximate phase field that is compatible with $d\phi$ s in a least-squares sense. We represent phases as unit complex numbers $\psi = e^{i\phi}$. Treating phases as unit complex numbers simplifies the problem as it removes the periodic discontinuities, allowing us to use gradient-based optimization methods. Then, along each edge ij of the mesh, we would like the phase difference between vertices i and j to be close to the integrated phase gradient: $\psi_j \approx \psi_i e^{i(d\phi \cdot \mathbf{e}_{ij})}$, where $\mathbf{e}_{ij}$ is the edge vector in reference space, and $d\phi$ is the wrinkle frequency averaged from adjacent faces. We define the compatibility energy for an edge as $E_{ij} = |\psi_j - \psi_i e^{i\omega_{ij}}|^2$, where $\omega_{ij} = d\phi \cdot \mathbf{e}_{ij}$ is the desired phase difference along the edge.

We incorporate some dynamics into the phases as well, which enables a temporally coherent evolution of the phase distribution. Specifically, we include both a physics-inspired potential energy estimating the contribution of phase reconstruction error to the elastic energy, and an inertia term modeling the kinetic energy associated with phase changes.

Recall that the wrinkle frequencies affect the elastic energy of the sheet via a term that scales as $\frac{h}{4} \|\frac{1}{2} a^2 d\phi d\phi^T\|_{SV}^2$, integrated over the area of the sheet. Analogously, we define the potential energy associated with the phase reconstruction error using a similar scaling,

$$E = \sum_{\text{edges } i,j} \frac{\beta h}{4} \left(\frac{a_i + a_j}{2} \right)^2 \frac{|\psi_i e^{i\omega_{ij}} - \psi_j|^2}{\|\mathbf{e}_{ij}\|^2} A_{ij}, \quad (12)$$

where A_{ij} is the area attributed to the edge ij , i.e. $\frac{1}{3}$ times the sum of the adjacent face areas, and β is the shear modulus of the material.

From the coarse-grained kinetic energy (8), we can also obtain a per-vertex inertia $T_i = \frac{1}{4} m_i a_i \dot{\phi}_i^2$ associated with the rate of change of phase at a vertex. Accordingly, we adopt a variational time integration strategy, where we compute the new phases by minimizing the sum of the potential energy E and an inertial term

$$T = \sum_{\text{vertices } i} \frac{m_i a_i^2}{4\Delta t^2} |\psi_i^{t+1} - \tilde{\psi}_i|^2. \quad (13)$$

Here $\tilde{\psi}_i$ is the predictor phase extrapolated from the previous two time steps, $\tilde{\psi}_i = \psi_i^t (\psi_i^t / \psi_i^{t-1})^{1-c\Delta t}$, incorporating a kinematic damping coefficient c . We perform the minimization by updating the phases at each vertex in Gauss-Seidel fashion.

Finally, the reconstructed phase values and wrinkle frequencies along the edges are then passed to the wrinkle up-sampling framework of Chen et al. [CKSV23] for rendering. This requires a single phase difference ω_{ij} along each edge, which is interpolated from the frequency vectors $d\phi$ on the two adjacent faces. To handle the sign ambiguity, we treat the wrinkle frequencies as 2-vectors [KCPS13] when averaging them. First, frequency vectors on faces are averaged to vertices, then the vectors on adjacent vertices are averaged and integrated to obtain phase differences on edges.

5. Results

All experiments are run on a machine with a 13th Gen Intel(R) Core(TM) i7-13700 CPU and 32 GB RAM, on the Ubuntu 24.04 operating system. The results showcase the efficacy of our parametric wrinkling framework for dynamic simulation of cloth. We use spatial hashing for broad-phase collision detection [THM*03] and constrained particles sampled in the mesh for narrow-phase collision handling [MMCK14]. Table 2 contains the physical parameters used to set up our examples. Statistics and performance numbers for all examples are given in Table 1. Our results show significant speedups when compared to high resolution results.

Cloth drape over sphere. Figure 1 shows the draping behavior of cloth on a sphere. A $1\text{m} \times 1\text{m}$ cloth is allowed to fall under the influence of gravity on a sphere of radius 0.6m. It can be seen that our formulation (Figure 1 right) allows high-frequency wrinkling on coarse base meshes, showcasing effects that are absent in the naive StVK simulation (Figure 1 left). High-frequency wrinkles are

Table 1: Mesh resolution and computation time per frame. For our results, we report the timing breakdown between dynamic simulation and phase upsampling in the format total (dynamics + upsampling).

	High resolution StVK		Our method	
	#verts	time (ms)	#verts	time (ms)
Sphere drape (Fig. 1)	16641	1140	289	18.53 (1.53 + 17)
Sphere drape (Fig. 7)	25521	5864	1644	7.8 (3.8 + 4)
Cloth drape (Fig. 3)	16641	586	1089	13.8 (6.8 + 7)
Twisting cylinder (Fig. 4)	4160	840	289	5.2 (0.2 + 5)
Sleeve (Fig. 5)	15617	23183	1056	26.6 (9.6 + 17)
Skirt (Fig. 6)	17477	203433	1149	94 (84 + 10)

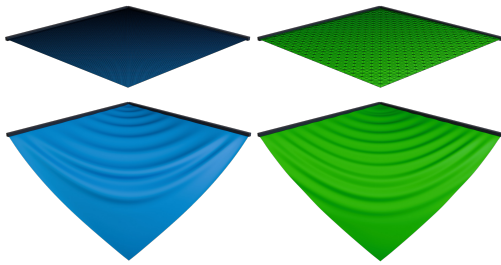


Figure 3: Cloth draping under gravity with two edges clamped. Left: High-resolution StVK simulation, right: our result.

visible in regions where the base mesh is compressed, and no wrinkling is visible in the top regions where the mesh is completely flat. The draping behavior adds visual plausibility to the simulation comparable to a reference result obtained using a high-resolution mesh with StVK energy (see Figure 1 middle).

Cloth drape with two ends fixed. Figure 3 shows cloth draping behavior when two edges of the cloth are clamped and the cloth is allowed to fall under gravity. Compression along the square cloth's diagonal introduces wrinkling in the mesh that is orthogonal to the compression. We compare our results with a high-resolution reference simulation (16641 vertices), and we observe that our method can capture the very high-frequency wrinkling behavior on a coarse mesh with just 1089 vertices. This provides a significant computational speedup compared to high-resolution simulation.

Cylinder twist. In Figure 4, we perform a cylinder-twist experiment to visualize dynamic wrinkle formation in a scene where one end of the cylinder is rotated. Vertices in the top ring of the cylinder are rotated anti-clockwise, keeping the vertices of the bottom ring fixed. The example shows wrinkle propagation from top to bottom as the cylinder is twisted.

Sleeve. In the real world, wrinkles commonly arise in garment regions such as shirt sleeves and trouser legs when the underlying body joints bend, inducing compression in the cloth sheet. To simulate this behavior, we set up a cloth sleeve draped over two cylinders joined via a ball in the middle. When the joint is bent, realistic buckling patterns can be seen in the simulation (Figure 5), although as discussed in Section 3, the diamond buckling pattern seen in the high-resolution reference cannot be reproduced.

Skirt on a dancing character. Finally, we test our method on a re-

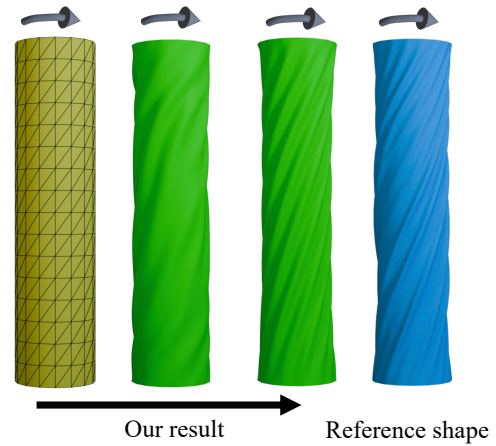


Figure 4: Twisting cloth on a cylinder. Far left: coarse base mesh used for simulation. Our results are shown from the first frame (middle left) to the last frame (middle right). Far right: high-resolution StVK simulation.

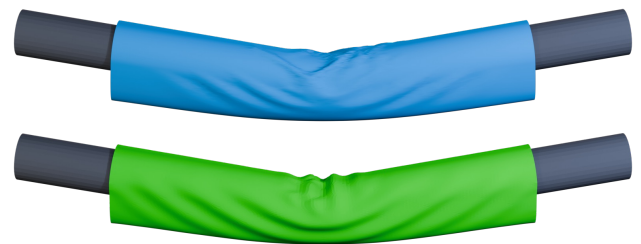
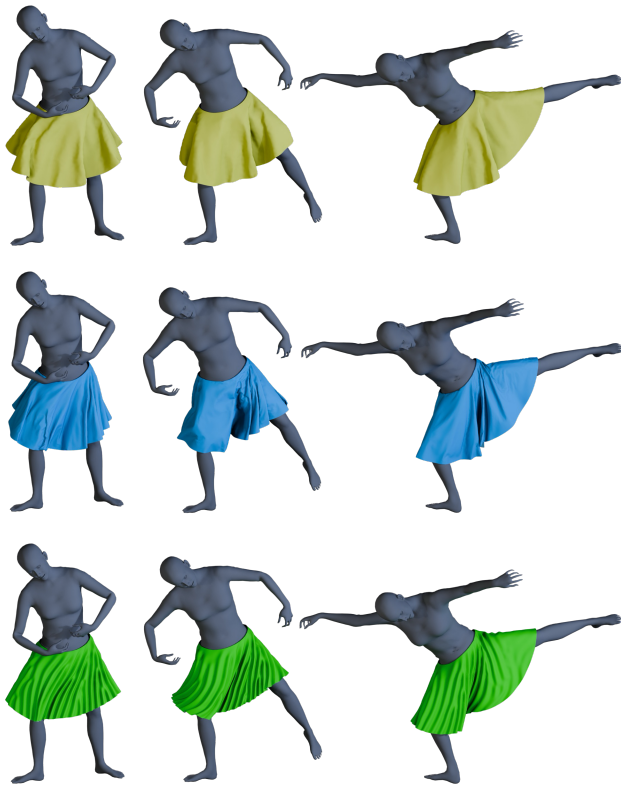


Figure 5: Wrinkling observed in the sleeve at the joint between two cylinders. Top: High resolution StVK simulation, bottom: our result.

alistic garment, by simulating a skirt on a motion capture sequence of a dancing human from the AMASS dataset [MGT*19]. Figure 6 shows the simulation frames. It can be seen that our method captures wrinkling in regions where none are present in the coarse base mesh. Our method supports wrinkle evolution, as visible in this example: wrinkles disappear in regions where the mesh flattens and appear in compressed regimes. The results show similar wrinkling amplitude and frequencies as a high-resolution reference

Table 2: Material properties and simulation parameters for cloth benchmark experiments.

Experiment	Y	ν	h	ρ	Δt (ms)
Sphere drape (Fig. 1)	2×10^4	0.5	1×10^{-3}	1.00×10^{-1}	0.5
Sphere drape (Fig. 7)	2×10^4	0.5	1×10^{-3}	1.00×10^{-1}	0.5
Cloth drape (Fig. 3)	1×10^4	0.5	1×10^{-3}	5.00×10^{-2}	1.25
Twisting cylinder (Fig. 4)	6×10^4	0.25	1×10^{-4}	2.76×10^{-2}	0.125
Sleeve (Fig. 5)	1.2×10^4	0.1	1×10^{-3}	2.76×10^{-2}	1
Skirt (Fig. 6)	5×10^5	0.5	1×10^{-4}	1.00×10^{-1}	0.21

**Figure 6:** Dynamic wrinkling on a dancing character. Top: high-frequency wrinkling is missing in the StVK simulation on the coarse base mesh. Middle: high-resolution StVK. Bottom: our method generates high-frequency wrinkles on top of a coarse simulation that respond to the character's motion.

in compressed regions. However, there are also noticeable differences, in particular some regions where our results are wrinkled whereas the reference simulation is smooth. *Effect of two-way coupling.* Our model incorporates a two-way coupling between the dynamics of base mesh and wrinkle parameters by design. To evaluate the benefit of this coupling, we compared our results with a simplified model in which the base mesh dynamics runs independently of the wrinkle parameters, then the wrinkle parameters are optimized using the sheet's potential energy (2). This strategy mimics the quasi-approach of [CCK*21]. Figure 7 shows the results, both when the standard StVK energy is used on the base mesh, and when

**Figure 7:** Comparison of different coupling models. Far left: high-resolution StVK simulation; middle left: wrinkle simulation without coupling from wrinkle parameters to coarse base mesh (StVK); middle right: wrinkle simulation without coupling from wrinkle parameters to coarse base mesh (TFT); far right: our result.

tension field theory is used (which removes compression resistance from StVK). With StVK, mesh artifacts due to insufficient resolution give rise to a bumpy, noisy appearance. With TFT, the base mesh hangs straight down since the wrinkles' contributions to the elastic energy are not accounted for. This shows the effectiveness of our coupled model, in which the base mesh captures the large-scale deformations, while small-scale details are captured by the wrinkle parameters.

Ablations on the effect of varying base mesh resolution. We further assess the robustness of our method against base mesh resolution. To this end, we run sphere drape simulations with base meshes having 289 vertices, 1089 vertices, and 4225 vertices respectively. The simulation results are shown in Figure 8. The results are similar qualitatively for all combinations of base-mesh resolutions. However, it should be noted that simulations with coarser resolution require more upsampling levels to obtain smooth results.

Ablations with varying phase damping coefficients. We assess the impact of phase dynamics by running the following evaluations on the Sphere drape example shown in Figure 1. First, we evaluate the effect of the inertial term (13) by running the phase reconstruction step using only the potential energy (12). In addition, we use the full phase reconstruction energy with a varying phase-damping coefficient (c) to assess the impact of damping. As the evaluation is purely dynamic and the outcomes cannot be captured in static images, we have included the results in the supplementary video. We state the observations here. Simulation without the inertial term produces sliding wrinkles throughout the cloth, breaking temporal coherence. Thus, an inertial term is required for obtaining temporally coherent wrinkling. Experiments with the damping coefficient reveal that, without damping $c = 0$, the wrinkles become too

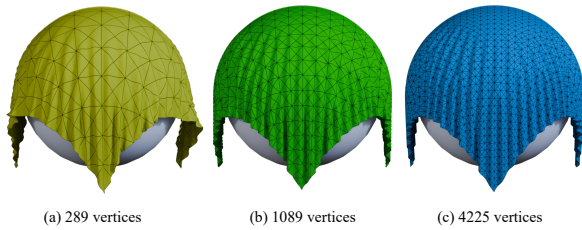


Figure 8: Experiments with varying base mesh resolutions, showing visually similar result.

dynamic, sliding across the cloth surface. Further, increasing the damping coefficient to moderate ($c = 15$) and very large ($c = 2000$) values yields visually good results, while maintaining temporal coherence in the simulations. Thus we safely choose $c = 15$ in all of our experiments.

Ablations with varying timestep sizes. To test how temporal discretization impacts our results, we ran an ablation study over varying time step sizes for the Cloth drape example shown in Figure 3. Since a proper evaluation requires dynamic visuals, we have included the results of this ablation study in the supplementary video. We highlight that the experiments show similar visual quality within the stable regime of the Symplectic Euler integrator (1.25 ms, 0.625 ms, and 0.3125 ms), whereas the largest time step that we tried (2.5 ms) leads to numerical instability.

6. Discussion and limitations

A unique feature of our approach is that the elastic response of the base mesh is coupled with the wrinkle parameters. While existing approaches such as inequality cloth [JLGF17] and tension field wrinkles [CCK*21] unconditionally remove all compression resistance from the elasticity model, in our model the compression resistance of the material decreases naturally with the emergence of wrinkles. This opens the door to allowing more realistic material models to be used, for example those estimated from observations [ZWL*25; MBT*12; CTT17] or computed by homogenization [SNW20]. It is unclear how to incorporate these in previous models, but our coarse-graining methodology can support these in principle. This is a promising avenue for future work, which would enable detailed simulation of more realistic fabric models in real time.

While the wrinkle parameters arising from our dynamics pipeline are temporally coherent (up to sign, in case of wrinkle frequencies), the phase upsampling framework sometimes produces jitter due to sign flips of the wrinkle frequency vector. We hope to address this in future work. Note that the method of [CKSV23] permits temporally coherent wrinkle evolution but only through a computationally expensive spacetime optimization, rendering it inapplicable to interactive simulations.

We were hopeful that the reduced compression resistance arising from wrinkle formation would eliminate locking artifacts in coarse simulations, but this did not turn out to be the case. For example, a square cloth mesh with all diagonals aligned in one direction ex-

hibits artificial resistance to bending in the perpendicular direction. We found that this issue arises because triangulations not aligned with the bending direction experience not just compression (along edges) but also tension (along triangle altitudes). The TFT model used by [CCK*21] suffers from the same locking problem, while inequality cloth [JLGF17] which uses only edge springs does not. To obtain coarse simulation results that are more consistent with the ground truth, this is an important problem that deserves further study.

As noted earlier, the wrinkle mesh formulation (1) is unable to model complex wrinkle patterns that cannot be approximated with spatially coherent sinusoids, for example in regions with biaxial compression, or interaction between compression and bending. In addition, our approach would not work well in cases where the base mesh resolution is insufficient to capture even the coarse behavior, such as interaction with sharp obstacles or narrow gaps.

There are several other potential extensions to the proposed technique that we hope to pursue in future work. First, since the wrinkle amplitudes are tracked as part of the base mesh simulation, it would be possible to incorporate them in the collision processing step as well, effectively treating the sheet as “thickened” by the wrinkle amplitude. This could even allow two-way coupling between wrinkles and contact, so that larger amplitudes could push the mid-surface away from obstacles, while regions in tight contact would be forced to have smaller amplitudes and higher frequencies. Secondly, we only have a rudimentary modeling of phase dynamics at present. It would be interesting to have a more sophisticated model including influences from external forces, so that for example the wrinkles in a flag could be pushed along by the wind.

Finally, we inherit from [CCK*21] the limitation that wrinkles are assumed to have locally a single frequency and direction: the wrinkle parameterization does not allow for superposition of multiple intersecting wrinkles, or more complex buckling patterns such as the diamond buckling seen in compressed, curved sheets. It may be possible to extend the model to allow for a superposition of multiple amplitude/frequency fields. More generally, an exciting possibility is to switch from the manually designed wrinkle parameterization to a learned representation that can span the range of local wrinkling and buckling patterns. Such a combination of coarse simulation and learned fine-scale details has been explored in previous works, but they do not achieve dynamic wrinkle behavior and do not incorporate sheet mechanics in the learned models. We are excited to explore this direction in future work.

7. Conclusion

We show that dynamic wrinkling is a feasible approach for detailed, physically consistent cloth animation on coarse meshes at interactive rates. Building on the wrinkle parameterization introduced by [CCK*21], we have introduced a dynamics model for a cloth sheet represented as a superposition of a base surface and a wrinkle distribution parameterized by spatially varying amplitude and frequency. Derived from Lagrangian mechanics by coarse-graining over wrinkle periods, our model naturally incorporates two-way coupling of the base surface and the wrinkle parameters, allowing the base deformation to drive emergence of wrinkles and permitting compres-

sion of the base surface in turn. To actually deform the mesh using the computed wrinkle parameters, we also introduce a simple phase reconstruction strategy that, combined with wrinkle upsampling [CKSV23], produces dynamically evolving, temporally coherent wrinkles on the simulated mesh.

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